

Fizika

Milliy sertifikat imtihoni topshiriqlari
30-sentabr 2026: shartlar, chizmalar va batafsil yechimlar

Online Fizika

Umumiy shart. Masalalarda berilmagan hollarda havo qarshiligi va yo'qotishlar hisobga olinmaydi, $g = 10 \text{ m/s}^2$.

Manba haqida. Topshiriqlar tartibi, variantlar va javoblar kaliti Asror Norqobilov (@ms_fizmat) to'plamiga mos keladi. Shartlar qayta bayon qilingan, chizma va grafiklar qaytadan chizilgan, yechimlar mustaqil tayyorlangan. Barcha javoblar manbadagi kalit bilan solishtirilgan va to'liq mos chiqdi.

A qism. 1–32-topshiriqlar (to'g'ri javobni belgilang)

1-masala

Jism 4 m/s tezlik bilan to'g'ri chiziqli tekis harakatlanmoqda. U 10 s da qancha yo'l bosib o'tadi?

A) $2,5 \text{ m}$ B) 14 m C) 40 m D) 400 m

Yechim. $S = vt = 4 \cdot 10 = 40 \text{ m}$.

Javob: C) 40 m

2-masala

Tinch holatdan tekis tezlanuvchan harakat boshlagan jismning dastlabki 3 s da bosib o'tgan yo'li harakatning oxirgi sekundida bosib o'tgan yo'lga teng. Harakat qancha vaqt davom etgan?

A) 5 s B) 4 s C) 6 s D) 9 s

Yechim. Tezlanishni a bilan belgilaymiz, umumiy harakat vaqti t bo'lsin. Dastlabki 3 s da yo'l:

$$s_3 = \frac{a \cdot 3^2}{2} = \frac{9a}{2}.$$

Oxirgi sekundda yo'l (jami yo'ldan dastlabki $t - 1 \text{ s}$ dagi yo'lni ayiramiz):

$$\frac{at^2}{2} - \frac{a(t-1)^2}{2} = a \left(t - \frac{1}{2} \right).$$

Ikkala yo'l teng:

$$\frac{9a}{2} = a \left(t - \frac{1}{2} \right) \Rightarrow t = \frac{9}{2} + \frac{1}{2} = 5 \text{ s}.$$

Javob: A) 5 s

3-masala

Jismning tezlanishini 4 marta orttirish uchun unga ta'sir etuvchi kuchni necha marta orttirish kerak?

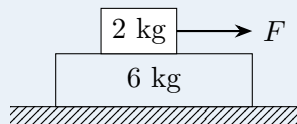
A) 2 marta B) 4 marta C) 8 marta D) 16 marta

Yechim. Nyutonning ikkinchi qonuni: $a = \frac{F}{m}$. Massa o'zgarmas bo'lgani uchun tezlanish kuchga to'g'ri proporsional, demak kuchni ham 4 marta orttirish kerak.

Javob: B) 4 marta

4-masala

Silliqlik gorizontali sirtida turgan 6 kg massali taxta ustiga 2 kg massali brusok qo'yilgan. Brusok va taxta orasidagi ishqalanish koeffitsiyenti 0,3. Brusokka gorizontali yo'nalishda 4 N kuch ta'sir qila boshladi (rasmga qarang). Taxtaning tezlanishini toping.



- A) 2 m/s^2 B) 1 m/s^2 C) $2/3 \text{ m/s}^2$ D) $0,5 \text{ m/s}^2$

Yechim. Avval jismlar sirpanishini tekshiramiz. Brusok va taxta orasidagi maksimal ishqalanish kuchi:

$$F_{ish,max} = \mu m_1 g = 0,3 \cdot 2 \cdot 10 = 6 \text{ N} > F = 4 \text{ N}.$$

Ishqalanish kuchi tashqi kuchdan katta, shuning uchun sirpanish bo'lmaydi: brusok va taxta bir butun bo'lib harakatlanadi.

$$a = \frac{F}{m_1 + m_2} = \frac{4}{2 + 6} = 0,5 \text{ m/s}^2.$$

Javob: D) $0,5 \text{ m/s}^2$

5-masala

Massasi 4 kg bo'lgan sun'iy yo'ldosh Yer sirtiga yaqin doiraviy orbita bo'ylab aylanmoqda. Yo'ldoshning kinetik energiyasini toping. Yer radiusi 6400 km.

- A) $1,28 \cdot 10^8 \text{ J}$ B) $2,56 \cdot 10^8 \text{ J}$ C) $6,4 \cdot 10^7 \text{ J}$ D) $1,28 \cdot 10^5 \text{ J}$

Yechim. Orbita Yer sirtiga yaqin, shuning uchun orbita radiusi $R = 6,4 \times 10^6 \text{ m}$ va markazga intilma tezlanish g ga teng:

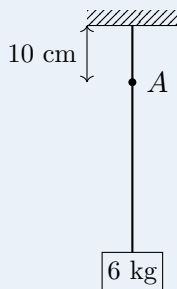
$$\frac{v^2}{R} = g \Rightarrow v = \sqrt{gR} = \sqrt{10 \cdot 6,4 \cdot 10^6} = 8000 \text{ m/s}.$$

$$E_k = \frac{mv^2}{2} = \frac{4 \cdot 6,4 \cdot 10^7}{2} = 1,28 \cdot 10^8 \text{ J}.$$

Javob: A) $1,28 \cdot 10^8 \text{ J}$

6-masala

Uzunligi 40 cm, massasi 4 kg bo'lgan bir jinsli arqonning yuqori uchi shipga mahkamlangan, pastki uchiga esa 6 kg massali yuk osilgan (rasmga qarang). Arqonning yuqori uchidan 10 cm pastda joylashgan A nuqtadagi elastiklik kuchini toping.



- A) 60 N B) 70 N C) 90 N D) 100 N

Yechim. A nuqtadagi kesim o'zidan pastdagi hamma narsaning og'irligini ushlab turadi: 6 kg yuk va arqonning pastki 30 cm qismi. Arqon bir jinsli, shuning uchun bu qismning massasi

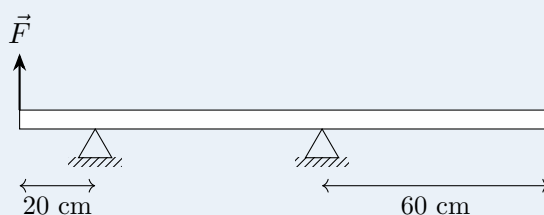
$$m_a = 4 \cdot \frac{30}{40} = 3 \text{ kg.}$$

$$F = (6 + 3)g = 9 \cdot 10 = 90 \text{ N.}$$

Javob: C) 90 N

7-masala

Uzunligi 140 cm, massasi 1200 kg bo'lgan bir jinsli balka ikkita tayanchda yotibdi. Tayanchlardan biri balkaning chap uchidan 20 cm, ikkinchisi o'ng uchidan 60 cm masofada joylashgan (rasmga qarang). Balkaning chap uchini ko'tarish uchun unga vertikal yuqoriga qo'yilishi kerak bo'lgan eng kichik kuchni toping.



A) 3 kN B) 1,5 kN C) 6 kN D) 0,75 kN

Yechim. Tayanchlar chap uchdan 20 cm va $140 - 60 = 80$ cm masofada. Og'irlik kuchi massa markazida (70 cm), $mg = 12000 \text{ N}$.

Chap uchni ko'targanda balka birinchi tayanchdan ajraladi va ikkinchi tayanch (80 cm) atrofida burila boshlaydi. Eng kichik kuch shu burilish boshlanish chegarasiga mos keladi: momentlar qoidasi (ikkinchi tayanchga nisbatan):

$$F \cdot 0,8 = mg \cdot 0,1 \Rightarrow F = \frac{12000 \cdot 0,1}{0,8} = 1500 \text{ N} = 1,5 \text{ kN.}$$

Javob: B) 1,5 kN

8-masala

Qirradi 4 m bo'lgan kub shaklidagi yopiq idish suv bilan to'la to'ldirilgan. Idish gorizontaal yo'nalishda $1,5 \text{ m/s}^2$ tezlanish bilan harakatlanmoqda. Idish devorlari suvga gorizontaal yo'nalishda qanday natijaviy kuch bilan ta'sir qiladi?

A) 9,6 kN B) 640 kN C) 24 kN D) 96 kN

Yechim. Suvga gorizontaal yo'nalishda faqat idish devorlari ta'sir qiladi (og'irlik kuchi vertikal). Suv idish bilan birga a tezlanish bilan harakatlanadi, shuning uchun devorlarning natijaviy gorizontaal kuchi Nyutonning ikkinchi qonuniga ko'ra ma ga teng.

Suv massasi: $m = \rho V = 1000 \cdot 4^3 = 64000 \text{ kg}$.

$$F = ma = 64000 \cdot 1,5 = 96000 \text{ N} = 96 \text{ kN.}$$

Javob: D) 96 kN

9-masala

Bikrligi 400 N/m bo'lgan prujinaga 0,16 kg massali yuk osilgan. Prujinali mayatnikning tebranish davrini toping ($\pi = 3,14$).

A) 0,0628 s B) 0,02 s C) 0,1256 s D) 1,256 s

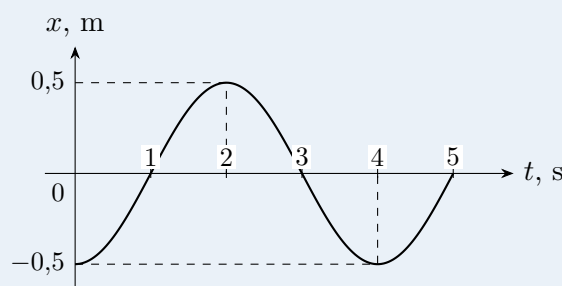
Yechim.

$$T = 2\pi\sqrt{\frac{m}{k}} = 2 \cdot 3,14 \cdot \sqrt{\frac{0,16}{400}} = 6,28 \cdot 0,02 = 0,1256 \text{ s.}$$

Javob: C) 0,1256 s

10-masala

Grafikda garmonik tebranayotgan jism koordinatasining vaqtga bog'lanishi berilgan. $t = 1$ s paytda jismning tezligini toping.



A) $\pi/4$ m/s B) $\pi/2$ m/s C) $\pi/8$ m/s D) 0

Yechim. Grafikdan: amplituda $A = 0,5$ m. Koordinata $t = 1, 3, 5$ s da nolga teng, ketma-ket ikki nol orasi yarim davr: $\frac{T}{2} = 2$ s, demak $T = 4$ s va

$$\omega = \frac{2\pi}{T} = \frac{\pi}{2} \text{ rad/s.}$$

Grafik $t = 0$ da $x = -A$ dan boshlanadi, $x = -A \cos \omega t$. Tezlik

$$v = x' = A\omega \sin \omega t.$$

$t = 1$ s da jism muvozanat vaziyatidan o'tmoqda ($\sin \frac{\pi}{2} = 1$), tezlik maksimal:

$$v = A\omega = 0,5 \cdot \frac{\pi}{2} = \frac{\pi}{4} \text{ m/s.}$$

Javob: A) $\pi/4$ m/s

11-masala

Geliy molekularining konsentratsiyasi $2,5 \cdot 10^{25} \text{ m}^{-3}$, o'rtacha kvadratik tezligi 600 m/s. Gaz bosimini toping ($M = 4 \text{ g/mol}$, $N_A = 6 \times 10^{23} \text{ mol}^{-1}$).

A) 60 kPa B) 10 kPa C) 40 kPa D) 20 kPa

Yechim. Bitta molekula massasi:

$$m_0 = \frac{M}{N_A} = \frac{4 \cdot 10^{-3}}{6 \cdot 10^{23}} \approx 6,67 \cdot 10^{-27} \text{ kg.}$$

MKT asosiy tenglamasi:

$$p = \frac{1}{3} n m_0 v^2 = \frac{1}{3} \cdot 2,5 \cdot 10^{25} \cdot 6,67 \cdot 10^{-27} \cdot 3,6 \cdot 10^5 = 2 \cdot 10^4 \text{ Pa} = 20 \text{ kPa.}$$

Javob: D) 20 kPa

12-masala

Zichligi $0,9 \text{ kg/m}^3$ bo'lgan gazning bosimi 108 kPa. Gaz molekularining o'rtacha kvadratik tezligini toping.

A) 346 m/s B) 600 m/s C) 1200 m/s D) 200 m/s

Yechim. $p = \frac{1}{3}\rho v^2$ dan

$$v = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3 \cdot 108000}{0,9}} = \sqrt{360000} = 600 \text{ m/s}.$$

Javob: B) 600 m/s

13-masala

8 g geliy o'zgarmas bosimda 831 J issiqlik miqdori oldi va uning temperaturasi 44°C ga yetdi. Gazning boshlang'ich temperaturasi toping ($R = 8,31 \text{ J/(mol} \cdot \text{K)}$).

A) 24°C B) $10,7^\circ\text{C}$ C) -6°C D) 64°C

Yechim. $\nu = \frac{8}{4} = 2 \text{ mol}$. Bir atomli gaz izobarik isiganda $Q = \frac{5}{2}\nu R\Delta T$:

$$\Delta T = \frac{2Q}{5\nu R} = \frac{2 \cdot 831}{5 \cdot 2 \cdot 8,31} = 20 \text{ K}.$$

Temperatura farqi Selsiy shkalasida ham 20° , demak

$$t_1 = 44 - 20 = 24^\circ\text{C}.$$

Javob: A) 24°C

14-masala

Temperaturasi -8°C bo'lgan 12 g muzni 20°C li suvga aylantirish uchun qancha issiqlik miqdori kerak? $c_{muz} = 2100 \text{ J/(kg} \cdot \text{K)}$, $\lambda = 330 \text{ kJ/kg}$, $c_{suv} = 4200 \text{ J/(kg} \cdot \text{K)}$.

A) 4,97 kJ B) 1,21 kJ C) 5,17 kJ D) 3,96 kJ

Yechim. $m = 0,012 \text{ kg}$. Jarayon uch bosqichdan iborat:

$$Q_1 = c_{muz}m \cdot 8 = 2100 \cdot 0,012 \cdot 8 = 201,6 \text{ J} \quad (\text{muzni } 0^\circ\text{C gacha isitish}),$$

$$Q_2 = \lambda m = 330000 \cdot 0,012 = 3960 \text{ J} \quad (\text{erish}),$$

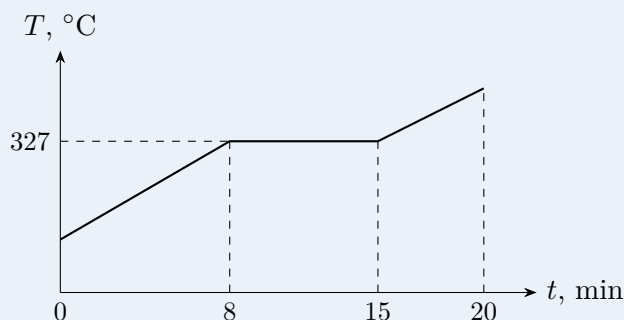
$$Q_3 = c_{suv}m \cdot 20 = 4200 \cdot 0,012 \cdot 20 = 1008 \text{ J} \quad (\text{suvni } 20^\circ\text{C gacha isitish}).$$

$$Q = 201,6 + 3960 + 1008 = 5169,6 \text{ J} \approx 5,17 \text{ kJ}.$$

Javob: C) 5,17 kJ

15-masala

Grafikda kristall modda temperaturasining qizdirish vaqtiga bog'lanishi berilgan. Moddaning erish jarayoni qancha vaqt davom etgan?



- A) 8 min B) 12 min C) 5 min D) 7 min

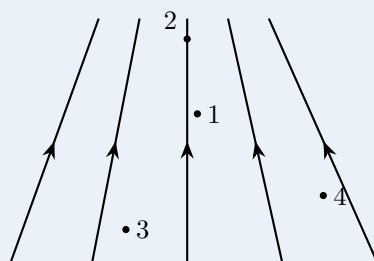
Yechim. Kristall modda erish vaqtida temperatura o'zgarmaydi: bu grafikdagi gorizontaal qism ($T = 327^\circ\text{C}$). U 8-minutda boshlanib 15-minutda tugaydi:

$$\Delta t = 15 - 8 = 7 \text{ min.}$$

Javob: D) 7 min

16-masala

Rasmda elektr maydonning kuch chiziqlari tasvirlangan. Qaysi nuqtada maydon kuchlanganligi eng katta?



- A) 1 B) 2 C) 3 D) 4

Yechim. Kuch chiziqlarining zichligi maydon kuchlanganligining kattaligini ko'rsatadi: chiziqlar qancha zich bo'lsa, E shuncha katta. Rasmda chiziqlar tepaga qarab yaqinlashadi, eng zich joy tepada joylashgan 2-nuqta.

Javob: B) 2

17-masala

Radiuslari 2 cm va 3 cm bo'lgan metall sharlarning zaryadlari mos ravishda $-5q$ va $+20q$. Bir-biridan uzoqda joylashgan bu sharlar ingichka o'tkazgich bilan tutashtirildi. O'tkazgich orqali qancha zaryad o'tadi?

- A) $11q$ B) $6q$ C) $9q$ D) $15q$

Yechim. Umumiy zaryad saqlanadi: $Q = -5q + 20q = 15q$. Tutashtirilganda sharlar potentsiali tenglashadi, $\varphi = \frac{kq}{r}$ bo'lgani uchun zaryad radiusga proporsional taqsimlanadi ($r_1 : r_2 = 2 : 3$):

$$q'_1 = 15q \cdot \frac{2}{5} = 6q, \quad q'_2 = 15q \cdot \frac{3}{5} = 9q.$$

Birinchi sharning zaryadi $-5q$ dan $+6q$ gacha o'zgardi, o'tkazgichdan o'tgan zaryad:

$$\Delta q = 6q - (-5q) = 11q.$$

Javob: A) $11q$

18-masala

Radiusi R bo'lgan metall sharning sirtiy zaryad zichligi $54,6 \text{ nC/m}^2$. Shar sirtidan R masofada joylashgan nuqtadagi elektr maydon kuchlanganligini toping ($\varepsilon_0 = 8,85 \times 10^{-12} \text{ F/m}$).

A) $6,17 \text{ kV/m}$ B) $3,08 \text{ kV/m}$ C) $1,54 \text{ kV/m}$ D) $0,77 \text{ kV/m}$

Yechim. Nuqta shar markazidan $r = 2R$ masofada. Shar tashqarisida maydon nuqtaviy zaryad maydoniga teng, $Q = \sigma \cdot 4\pi R^2$:

$$E = \frac{Q}{4\pi\varepsilon_0(2R)^2} = \frac{\sigma \cdot 4\pi R^2}{4\pi\varepsilon_0 \cdot 4R^2} = \frac{\sigma}{4\varepsilon_0}.$$

$$E = \frac{54,6 \cdot 10^{-9}}{4 \cdot 8,85 \cdot 10^{-12}} \approx 1,54 \cdot 10^3 \text{ V/m} = 1,54 \text{ kV/m}.$$

Javob: C) $1,54 \text{ kV/m}$

19-masala

Jadvalda rezistorlar to'plami berilgan. Barcha rezistorlardan foydalanib hosil qilish mumkin bo'lgan eng kichik qarshilikni toping.

Rezistor	Qarshiligi	Soni
Qizil	5Ω	2 ta
Yashil	15Ω	3 ta
Ko'k	25Ω	5 ta

A) 2Ω B) $0,8 \Omega$ C) $1,5 \Omega$ D) $1,25 \Omega$

Yechim. Eng kichik qarshilik barcha rezistorlarni parallel ulaganda hosil bo'ladi:

$$\frac{1}{R} = \frac{2}{5} + \frac{3}{15} + \frac{5}{25} = 0,4 + 0,2 + 0,2 = 0,8 \Rightarrow R_{min} = \frac{1}{0,8} = 1,25 \Omega.$$

Javob: D) $1,25 \Omega$

20-masala

Atomda elektron yadro atrofida radiusi R bo'lgan aylana bo'ylab harakatlanmoqda (yadro zaryadi $+e$). Elektronning orbita bo'ylab harakati hosil qiladigan tok kuchini toping (k – Kulon qonunidagi koeffitsiyent, m_e – elektron massasi).

A) $\frac{e}{2\pi R} \sqrt{\frac{k}{m_e R}}$ B) $\frac{e^2}{2\pi R} \sqrt{\frac{k}{m_e R}}$ C) $\frac{e^2}{2R} \sqrt{\frac{k}{m_e R}}$ D) $\frac{e^2}{\pi R} \sqrt{\frac{k}{m_e R}}$

Yechim. Kulon kuchi markazga intilma kuch vazifasini bajaradi:

$$\frac{ke^2}{R^2} = \frac{m_e v^2}{R} \Rightarrow v = e \sqrt{\frac{k}{m_e R}}.$$

Aylanish davri $T = \frac{2\pi R}{v}$, tok kuchi $I = \frac{e}{T}$:

$$I = \frac{ev}{2\pi R} = \frac{e^2}{2\pi R} \sqrt{\frac{k}{m_e R}}.$$

Javob: B) $\frac{e^2}{2\pi R} \sqrt{\frac{k}{m_e R}}$

21-masala

Al^{3+} ionlari bo'lgan eritmadan $5 \cdot 96500$ C zaryad o'tdi. Katodda qancha modda miqdori alyuminiy ajraladi? $F = 96\,500$ C/mol.

A) 5 mol B) 15 mol C) 1,67 mol D) 0,6 mol

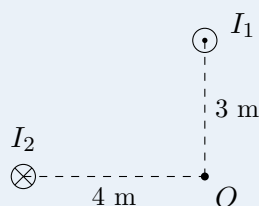
Yechim. Faradey qonuni: bir mol modda ajralishi uchun zF zaryad kerak ($z = 3$):

$$\nu = \frac{q}{zF} = \frac{5 \cdot 96500}{3 \cdot 96500} = \frac{5}{3} \approx 1,67 \text{ mol}.$$

Javob: C) 1,67 mol

22-masala

Ikkita uzun to'g'ri o'tkazgich chizma tekisligiga tik joylashgan va ulardan bir xil tok o'tmoqda (rasmga qarang). Birinchi o'tkazgichning O nuqtada hosil qilgan magnit maydon induksiyasi 120 mT. O nuqtadagi natijaviy magnit induksiyaning toping.



A) 150 mT B) 210 mT C) 30 mT D) 90 mT

Yechim. Toklar teng, to'g'ri tokning maydoni $B = \frac{\mu_0 I}{2\pi r} \propto \frac{1}{r}$. I_1 dan O gacha $r_1 = 3$ m, I_2 dan O gacha $r_2 = 4$ m:

$$B_2 = B_1 \frac{r_1}{r_2} = 120 \cdot \frac{3}{4} = 90 \text{ mT}.$$

Yo'nalishlar (o'ng qo'l qoidasi): I_1 (O dan yuqorida, tok chizmadan tomon) maydoni O da gorizontaal (o'ngga), I_2 (O dan chapda, tok chizmaga tomon) maydoni vertikal (pastga). Demak $\vec{B}_1 \perp \vec{B}_2$:

$$B = \sqrt{B_1^2 + B_2^2} = \sqrt{120^2 + 90^2} = 150 \text{ mT}.$$

Javob: A) 150 mT

23-masala

Induktivligi 5 mH bo'lgan g'altakdagi tok kuchi $I = 2,4 + 1,6t$ (SI da) qonun bo'yicha o'zgaradi. G'altakda hosil bo'ladigan o'zinduksiya EYuK ning modulini toping.

A) 12 mV B) 20 mV C) 3,2 mV D) 8 mV

Yechim. Tok o'zgarish tezligi $\frac{\Delta I}{\Delta t} = 1,6 \text{ A/s}$:

$$|\mathcal{E}| = L \left| \frac{\Delta I}{\Delta t} \right| = 5 \cdot 10^{-3} \cdot 1,6 = 8 \cdot 10^{-3} \text{ V} = 8 \text{ mV}.$$

Javob: D) 8 mV

24-masala

G'altakdan 5 A tok o'tganda undagi magnit oqimi 0,4 Wb ga teng. G'altakdagi magnit maydon energiyasini toping.

A) 2 J B) 1 J C) 0,5 J D) 0,08 J

Yechim. $\Phi = LI$ dan $L = \frac{\Phi}{I} = \frac{0,4}{5} = 0,08 \text{ H}$. Energiya:

$$W = \frac{LI^2}{2} = \frac{\Phi I}{2} = \frac{0,4 \cdot 5}{2} = 1 \text{ J}.$$

Javob: B) 1 J

25-masala

Tebranish konturidagi kondensatorning sig'imi 25 nF, g'altakning induktivligi 1 mH. Kontur nurlaydigan elektromagnit to'lqin uzunligini toping ($\pi \approx 3$, $c = 3 \times 10^8 \text{ m/s}$).

A) 9 km B) 1,5 km C) 4,5 km D) 18 km

Yechim. Kontur davri (Tomson formulasi):

$$T = 2\pi\sqrt{LC} = 2 \cdot 3 \cdot \sqrt{10^{-3} \cdot 25 \cdot 10^{-9}} = 6 \cdot \sqrt{2,5 \cdot 10^{-11}} = 6 \cdot 5 \cdot 10^{-6} = 3 \times 10^{-5} \text{ s}.$$

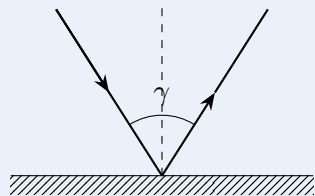
To'lqin uzunligi:

$$\lambda = cT = 3 \cdot 10^8 \cdot 3 \cdot 10^{-5} = 9000 \text{ m} = 9 \text{ km}.$$

Javob: A) 9 km

26-masala

Yorug'lik nuri yassi ko'zgudan qaytmoqda. Tushgan va qaytgan nurlar orasidagi burchak γ ga teng (rasmga qarang). Tushish burchagi qanday?



A) γ B) 2γ C) $90^\circ - \gamma/2$ D) $\gamma/2$

Yechim. Qaytish qonuni: tushish burchagi qaytish burchagiga teng, $\varphi_{tush} = \varphi_{qayt}$ (ikkalasi normaldan o'lchanadi). Tushgan va qaytgan nurlar orasidagi burchak shu ikki burchak yig'indisiga teng:

$$\gamma = \varphi_{tush} + \varphi_{qayt} = 2\varphi_{tush} \Rightarrow \varphi_{tush} = \frac{\gamma}{2}.$$

Javob: D) $\gamma/2$

27-masala

Fokus masofalari 15 cm va 20 cm bo'lgan ikkita yig'uvchi linza bir-biridan 11 cm masofada umumiy bosh optik o'q bo'yicha joylashgan. Linzalar sistemasining optik kuchini toping (dptr).

A) 4 B) 5 C) 8 D) 12

Yechim. Linzalar orasidagi masofa l bo'lganda sistemaning optik kuchi $D = D_1 + D_2 - lD_1D_2$:

$$D_1 = \frac{1}{0,15} = \frac{20}{3}, \quad D_2 = \frac{1}{0,2} = 5,$$

$$D = \frac{20}{3} + 5 - 0,11 \cdot \frac{20}{3} \cdot 5 = \frac{35}{3} - \frac{11}{3} = \frac{24}{3} = 8 \text{ dptr}.$$

Javob: C) 8 dptr

28-masala

Jismning to'liq energiyasi uning tinchlikdagi energiyasidan 4 marta katta. Jismning tezligini toping (c – yorug'likning vakuumdagi tezligi).

A) $\frac{\sqrt{3}}{2}c$ B) $\frac{\sqrt{15}}{4}c$ C) $0,75c$ D) $\frac{c}{4}$

Yechim. $E = \gamma E_0$, bundan $\gamma = 4$:

$$\frac{1}{\sqrt{1 - v^2/c^2}} = 4 \Rightarrow 1 - \frac{v^2}{c^2} = \frac{1}{16} \Rightarrow v = c\sqrt{\frac{15}{16}} = \frac{\sqrt{15}}{4}c.$$

Javob: B) $\frac{\sqrt{15}}{4}c$

29-masala

${}_{47}^{108}\text{X}$ yadrosidagi neytronlar sonini aniqlang.

A) 47 B) 108 C) 155 D) 61

Yechim. $N = A - Z = 108 - 47 = 61$.

Javob: D) 61

30-masala

Radioaktiv izotopning yarim yemirilish davri T . $4T$ vaqt o'tgach, yemirilmagan yadrolar massasi boshlang'ichiga nisbatan necha marta kamayadi?

A) 16 marta B) 4 marta C) 8 marta D) 32 marta

Yechim. $m = m_0 \cdot 2^{-t/T} = m_0 \cdot 2^{-4} = \frac{m_0}{16}$.

Javob: A) 16 marta

31-masala

Tinch turgan elektron va pozitron annigilyatsiyasi natijasida ikkita bir xil γ -kvant hosil bo'ldi. Har bir γ -kvantning energiyasini toping ($m_e = 9,1 \times 10^{-31}$ kg, $c = 3 \times 10^8$ m/s).

A) $1,64 \cdot 10^{-13}$ J B) $4,1 \cdot 10^{-14}$ J C) $8,19 \cdot 10^{-14}$ J D) $2,73 \cdot 10^{-22}$ J

Yechim. Annigilyatsiyada elektron va pozitron massasi energiyaga aylanadi: $2m_e c^2$. Ikkita bir xil kvant hosil bo'lgani uchun

$$E = \frac{2m_e c^2}{2} = m_e c^2 = 9,1 \cdot 10^{-31} \cdot 9 \cdot 10^{16} = 8,19 \cdot 10^{-14} \text{ J.}$$

Javob: C) $8,19 \cdot 10^{-14} \text{ J}$

32-masala

Quvvati 2 kW bo'lgan konditsioner har kuni 5 soatdan ishlatiladi. 1 kW · h elektr energiyasining narxi 600 so'm bo'lsa, 30 kunda konditsioner uchun qancha to'lov qilinadi?

A) 60 000 so'm B) 180 000 so'm C) 18 000 so'm D) 360 000 so'm

Yechim. 30 kunda sarflangan energiya: $W = Pt = 2 \cdot 5 \cdot 30 = 300 \text{ kW} \cdot \text{h}$. To'lov:

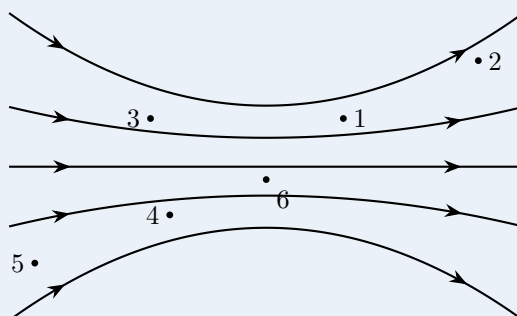
$$300 \cdot 600 = 180\,000 \text{ so'm.}$$

Javob: B) 180 000 so'm

B qism. 33–35-topshiriqlar (moslashtirish)

33–35-masala

Rasmda elektr maydonning kuch chiziqlari va 1–6 nuqtalar tasvirlangan.



A) 1 B) 2 C) 3 D) 4 E) 5 F) 6

33. Potensial eng kichik bo'lgan nuqta. 34. Elektr maydon kuchlanganligi eng katta bo'lgan nuqta. 35. Potensial eng katta bo'lgan nuqta.

Yechim. Maydon yo'nalishi bo'yicha (kuch chiziqlari bo'ylab, chapdan o'ngga) potensial kamayadi.

33. Eng o'ngda joylashgan 2-nuqtada potensial eng kichik $\Rightarrow$ **B**.

34. Kuch chiziqlari eng zich bo'lgan joy chiziqlar o'rtada yaqinlashgan 6-nuqta $\Rightarrow$ **F**.

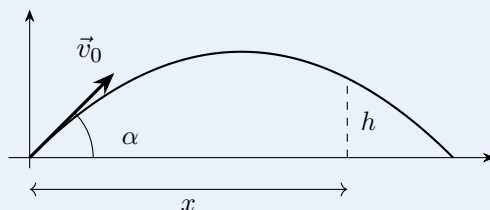
35. Eng chapda joylashgan 5-nuqtada potensial eng katta $\Rightarrow$ **E**.

Javob: 33 – B; 34 – F; 35 – E

C qism. 36–45-topshiriqlar (javobni yozing)

36-masala

Jism gorizontalga 45° burchak ostida 20 m/s boshlang'ich tezlik bilan otildi (rasmga qarang).



- a) Jism qaytib tushayotganda yerdan $h = 7,5 \text{ m}$ balandlikda bo'ladigan nuqta otilgan joydan gorizontal bo'yicha qanday x masofada joylashgan?
 b) Jism ko'tarilgan maksimal balandlikni toping.

Yechim. a) Traektoriya tenglamasi:

$$y = x \tan \alpha - \frac{gx^2}{2v_0^2 \cos^2 \alpha} = x - \frac{10x^2}{2 \cdot 400 \cdot 0,5} = x - \frac{x^2}{40}.$$

$y = 7,5$ bo'lganda:

$$x - \frac{x^2}{40} = 7,5 \Rightarrow x^2 - 40x + 300 = 0 \Rightarrow x = 10 \text{ yoki } x = 30 \text{ m.}$$

$x = 10 \text{ m}$ jism ko'tarilayotgandagi nuqta, qaytib tushayotgandagi nuqta $x = 30 \text{ m}$.

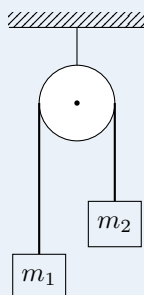
b)

$$H = \frac{v_0^2 \sin^2 \alpha}{2g} = \frac{400 \cdot 0,5}{20} = 10 \text{ m.}$$

Javob: a) $x = 30 \text{ m}$; b) $H = 10 \text{ m}$

37-masala

Vaznsiz blok orqali o'tkazilgan vaznsiz, cho'zilmas ipning uchlariga massalari $m_1 = 600 \text{ g}$ va $m_2 = 400 \text{ g}$ bo'lgan yuklar osilgan. Blokdagi ishqalanishni hisobga olmang.



- a) Yuklarning tezlanishini toping.
 b) Ipning taranglik kuchini toping.

Yechim. $m_1 > m_2$ bo'lgani uchun m_1 pastga, m_2 yuqoriga harakatlanadi. Har bir yuk uchun Nyutonning ikkinchi qonuni:

$$\begin{cases} m_1 g - T = m_1 a \\ T - m_2 g = m_2 a \end{cases}$$

a) Tenglamalarni qo'shamiz:

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2} = \frac{0,2 \cdot 10}{1} = 2 \text{ m/s}^2.$$

b)

$$T = m_2(g + a) = 0,4 \cdot 12 = 4,8 \text{ N}.$$

Javob: a) $a = 2 \text{ m/s}^2$; b) $T = 4,8 \text{ N}$

38-masala

Prujinaga 6 kg massali yuk osilganda uning uzunligi 35 cm bo'ldi. Yukning $\frac{1}{5}$ qismi olib tashlanganda prujina 3 cm ga qisqardi.

a) Prujinaning bikrligini toping.

b) Prujinaning deformatsiyalanmagan holatdagi uzunligini toping.

Yechim. a) Olib tashlangan massa: $\Delta m = \frac{6}{5} = 1,2 \text{ kg}$, uning og'irligi 12 N prujina kuchining kamayishiga teng:

$$k \Delta x = \Delta m g \Rightarrow k = \frac{12}{0,03} = 400 \text{ N/m}.$$

b) Dastlabki cho'zilish: $x_0 = \frac{mg}{k} = \frac{60}{400} = 0,15 \text{ m} = 15 \text{ cm}$. Cho'zilgan uzunlik 35 cm, demak

$$l_0 = 35 - 15 = 20 \text{ cm}.$$

Javob: a) $k = 400 \text{ N/m}$; b) $l_0 = 20 \text{ cm}$

39-masala

Hajmi 4 m^3 , temperaturasi 27°C bo'lgan ideal gaz 100 kPa o'zgarmas bosimda qizdirilib, temperaturasi 120 K ga oshirildi.

a) Gazning oxirgi hajmini toping.

b) Gaz bajargan ishni toping.

Yechim. Izobarik jarayon, $T_1 = 300 \text{ K}$, $T_2 = 300 + 120 = 420 \text{ K}$.

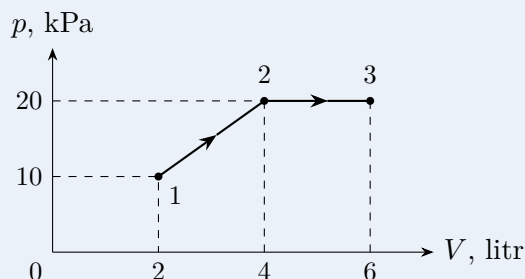
a) $\frac{V_2}{V_1} = \frac{T_2}{T_1} \Rightarrow V_2 = 4 \cdot \frac{420}{300} = 5,6 \text{ m}^3$.

b) $A = p \Delta V = 100 \cdot 10^3 \cdot (5,6 - 4) = 1,6 \times 10^5 \text{ J} = 160 \text{ kJ}$.

Javob: a) $V_2 = 5,6 \text{ m}^3$; b) $A = 160 \text{ kJ}$

40-masala

Bir atomli ideal gaz grafikda ko'rsatilgan $1 \rightarrow 2 \rightarrow 3$ jarayonni o'tdi.



a) Gaz bajargan umumiy ishni toping.

b) Gazga berilgan issiqlik miqdorini toping.

Yechim. Grafikdan: $p_1 = 10 \text{ kPa}$, $V_1 = 2 \text{ l}$; $p_2 = p_3 = 20 \text{ kPa}$, $V_2 = 4 \text{ l}$, $V_3 = 6 \text{ l}$. $1 \text{ kPa} \cdot \text{l} = 1 \text{ J}$.

a) Ish grafik ostidagi yuzaga teng:

$$A_{12} = \frac{10 + 20}{2} \cdot (4 - 2) = 30 \text{ J}, \quad A_{23} = 20 \cdot (6 - 4) = 40 \text{ J}, \quad A = 70 \text{ J}.$$

b) Bir atomli gaz uchun $\Delta U = \frac{3}{2}(p_3 V_3 - p_1 V_1)$:

$$p_3 V_3 = 120 \text{ J}, \quad p_1 V_1 = 20 \text{ J} \Rightarrow \Delta U = \frac{3}{2} \cdot 100 = 150 \text{ J}.$$

Termodinamikaning birinchi qonuni:

$$Q = \Delta U + A = 150 + 70 = 220 \text{ J}.$$

Javob: a) $A = 70 \text{ J}$; b) $Q = 220 \text{ J}$

41-masala

Diametri $2,2 \text{ m}$ bo'lgan yakkalangan metall shar berilgan ($k = 9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$).

a) Sharning elektr sig'imini toping.

b) Shar potentsiali 2700 V bo'lishi uchun unga qancha zaryad berish kerak?

Yechim. Radius $R = 1,1 \text{ m}$.

a) Yakkalangan shar sig'imi $C = 4\pi\epsilon_0 R = \frac{R}{k}$:

$$C = \frac{1,1}{9 \cdot 10^9} \approx 1,22 \cdot 10^{-10} \text{ F} = 122 \text{ pF}.$$

b) $q = C\varphi = \frac{R\varphi}{k} = \frac{1,1 \cdot 2700}{9 \cdot 10^9} = 3,3 \cdot 10^{-7} \text{ C}.$

Javob: a) $C \approx 1,22 \cdot 10^{-10} \text{ F}$ ($\approx 122 \text{ pF}$); b) $q = 3,3 \cdot 10^{-7} \text{ C}$

42-masala

Metall o'tkazgichning qarshiligi -20°C da 12Ω , 20°C da esa $12,6 \Omega$ ga teng.

a) Qarshilikning temperatura koeffitsiyentini toping.

b) 1000°C da o'tkazgichning qarshiligi qanday bo'ladi?

Yechim. Qonun: $R = R_0(1 + \alpha t)$, bunda $R_0 - 0^\circ\text{C}$ dagi qarshilik.

$$12 = R_0(1 - 20\alpha), \quad 12,6 = R_0(1 + 20\alpha).$$

Qo'shamiz: $2R_0 = 24,6 \Rightarrow R_0 = 12,3 \Omega$. Ayiramiz: $40\alpha R_0 = 0,6$:

$$\alpha = \frac{0,6}{40 \cdot 12,3} = \frac{1}{820} \approx 1,22 \cdot 10^{-3} \text{ K}^{-1}.$$

b)

$$R(1000) = 12,3 \left(1 + \frac{1000}{820} \right) = 12,3 + 15 = 27,3 \Omega.$$

Javob: a) $\alpha \approx 1,22 \cdot 10^{-3} \text{ K}^{-1}$; b) $R = 27,3 \Omega$

43-masala

Yig'uvchi linza buyumning 4 marta kattalashgan to'g'ri (mavhum) tasvirini hosil qiladi. Buyum bilan uning tasviri orasidagi masofa 30 cm.

- a) Buyumdan linzagacha bo'lgan masofani toping.
b) Linzaning optik kuchini toping.

Yechim. To'g'ri (mavhum) tasvir buyum bilan bir tomonda va undan uzoqroqda joylashadi: $f - d = 30$ cm, kattalashtirish $\Gamma = \frac{f}{d} = 4 \Rightarrow f = 4d$.

a) $4d - d = 30 \Rightarrow d = 10$ cm, $f = 40$ cm.

b) Mavhum tasvir uchun linza formulasi $\frac{1}{F} = \frac{1}{d} - \frac{1}{f}$:

$$D = \frac{1}{0,1} - \frac{1}{0,4} = 10 - 2,5 = 7,5 \text{ dptr.}$$

Javob: a) $d = 10$ cm; b) $D = 7,5$ dptr

44-masala

Tinchlikdagi massasi 6 kg, uzunligi 12 m bo'lgan sterjen o'z uzunligi bo'ylab harakatlanmoqda. Bunda uning massasi 4 kg ga ortdi ($c = 3 \times 10^8$ m/s).

- a) Harakatlanayotgan sterjenning uzunligini toping.
b) Sterjenning kinetik energiyasini toping.

Yechim. Harakatdagi massa $m = 6 + 4 = 10$ kg, demak $\gamma = \frac{m}{m_0} = \frac{10}{6} = \frac{5}{3}$.

a) Relyativistik uzunlik qisqarishi:

$$l = \frac{l_0}{\gamma} = 12 \cdot \frac{3}{5} = 7,2 \text{ m.}$$

b) $E_k = (m - m_0)c^2 = 4 \cdot 9 \cdot 10^{16} = 3,6 \cdot 10^{17}$ J.

Javob: a) $l = 7,2$ m; b) $E_k = 3,6 \cdot 10^{17}$ J

45-masala

Vodorod atomining ionlanish energiyasi 13,6 eV. Asosiy holatdagi atom foton yutishi natijasida elektronning impuls momenti 2 marta ortdi.

- a) Elektronning orbita bo'ylab aylanish davrlari nisbati T_2/T_1 ni toping.
b) Atom qancha energiyali foton yutgan?

Yechim. Bor postulatiga ko'ra impuls momenti $L = n\hbar$. Asosiy holatda $n = 1$; L ikki marta ortsa, $n = 2$: elektron ikkinchi sathga o'tdi.

a) Bor modelida $r_n \propto n^2$, $v_n \propto \frac{1}{n}$, shuning uchun

$$T_n = \frac{2\pi r_n}{v_n} \propto n^3 \Rightarrow \frac{T_2}{T_1} = 2^3 = 8.$$

b)

$$E_\gamma = E_2 - E_1 = 13,6 \left(1 - \frac{1}{4}\right) = 10,2 \text{ eV.}$$

Javob: a) $T_2/T_1 = 8$; b) $E_\gamma = 10,2$ eV

Javoblar kaliti

1–32-topshiriqlar

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
C	A	B	D	A	C	B	D	C	A	D	B	A	C	D	B
17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
A	C	D	B	C	A	D	B	A	D	C	B	D	A	C	B

33–35-topshiriqlar

33	34	35
B	F	E

36–45-topshiriqlar

T/r	a)	b)
36	30 m	10 m
37	2 m/s^2	4,8 N
38	400 N/m	20 cm
39	$5,6 \text{ m}^3$	160 kJ
40	70 J	220 J
41	$\approx 1,22 \cdot 10^{-10} \text{ F}$ ($\approx 122 \text{ pF}$)	$3,3 \cdot 10^{-7} \text{ C}$
42	$\approx 1,22 \cdot 10^{-3} \text{ K}^{-1}$	27,3 Ω
43	10 cm	7,5 dptr
44	7,2 m	$3,6 \cdot 10^{17} \text{ J}$
45	8	10,2 eV